

RYONGYON, Korea, Democratic People's Republic of

WMO: 470680

Lat: 38.150N

Lon: 124.883E

Elev: 20

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.2	12.7	0.2	5.6	15.8	3.6	6.6	17.6	21.1	22.2	19.7	22.7	4.5	320	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	87.8	78.0	85.4	76.4	83.5	75.2	80.0	85.6	78.7	83.4	77.5	81.6	5.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.3	147.2	83.6	77.3	142.0	82.0	76.2	137.0	80.8	43.7	86.1	42.3	83.8	41.1	81.8	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.0	17.5	15.0	2.8	91.0	5.2	1.8	-1.0	92.3	-4.0	93.4	-6.9	94.4	-10.7	95.8		
			2.9	82.4	4.7	2.4	-0.5	84.1	-3.2	85.5	-5.9	86.9	-9.3	88.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	25.4	29.7	39.3	51.0	61.3	70.1	76.0	76.8	68.0	56.4	42.8	30.1	(6)
(7)		DBStd	18.71	6.87	6.69	5.90	5.89	4.53	3.64	3.43	4.01	4.65	5.92	7.39	7.28	(7)
(8)		HDD50	2593	764	569	335	57	0	0	0	0	0	15	236	617	(8)
(9)		HDD65	5613	1229	989	797	419	130	6	0	0	24	270	667	1082	(9)
(10)		CDD50	3453	0	0	3	88	352	604	805	830	539	214	18	0	(10)
(11)		CDD65	1001	0	0	0	0	17	160	340	366	113	5	0	0	(11)
(12)		CDH74	6715	0	0	0	6	174	850	2207	2862	593	23	0	0	(12)
(13)		CDH80	1496	0	0	0	0	15	148	492	776	64	1	0	0	(13)
(14)	Wind	WSAvg	5.1	4.8	5.5	6.4	6.5	6.1	4.7	4.8	4.2	3.4	4.2	5.1	5.1	(14)
(15)	Precipitation	PrecAvg	34.0	0.5	0.7	0.7	2.0	3.1	3.7	9.6	7.3	3.6	1.4	1.5	0.8	(15)
(16)		PrecMax	66.3	1.3	2.4	2.0	4.4	10.0	10.9	20.0	17.6	17.0	3.9	4.8	1.9	(16)
(17)		PrecMin	17.6	0.0	0.0	0.0	0.1	0.2	0.5	1.8	1.1	0.2	0.0	0.2	0.1	(17)
(18)		PrecStd	10.1	0.4	0.6	0.6	1.0	2.2	2.5	4.8	4.5	3.8	1.1	1.1	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	51.2	61.8	74.0	82.0	86.7	90.5	91.4	84.8	77.0	64.7	51.3	(19)
(20)		MCWB	41.4	45.2	50.0	59.2	65.5	70.3	80.2	80.0	73.2	63.9	58.2	46.9	(20)	
(21)		2%	DB	41.5	46.1	56.8	68.8	78.4	83.8	87.1	88.7	81.8	73.0	61.0	47.5	(21)
(22)		MCWB	37.1	40.4	47.7	56.7	62.9	69.6	78.1	79.2	70.9	62.8	55.6	43.6	(22)	
(23)		5%	DB	38.5	43.2	53.3	65.7	75.4	81.3	84.7	86.3	79.7	70.2	58.0	44.0	(23)
(24)		MCWB	34.7	38.1	45.4	54.1	61.6	69.0	76.9	78.0	69.7	60.9	53.0	40.1	(24)	
(25)		10%	DB	35.7	40.3	49.9	62.4	72.4	78.9	82.7	84.3	77.7	67.4	55.0	41.1	(25)
(26)		MCWB	32.4	36.2	43.2	52.3	60.3	68.2	76.0	77.1	68.5	59.5	50.1	37.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	46.8	52.7	62.7	68.2	74.1	81.9	81.9	76.8	67.3	60.1	48.2	(27)
(28)		MADB	44.8	49.9	57.8	69.4	76.8	80.8	87.6	88.2	80.6	72.8	62.9	51.2	(28)	
(29)		2%	WB	37.7	41.8	49.4	59.2	65.7	72.6	80.0	80.4	73.7	64.7	56.6	44.0	(29)
(30)		MADB	40.6	45.1	54.7	66.2	74.2	78.7	85.3	86.4	78.6	70.3	59.6	46.6	(30)	
(31)		5%	WB	35.1	38.9	46.6	56.6	63.9	71.3	78.4	79.2	71.9	62.8	53.6	40.5	(31)
(32)		MADB	37.9	42.3	52.3	63.0	71.5	77.5	82.8	84.4	77.4	67.9	57.6	43.5	(32)	
(33)		10%	WB	32.7	36.4	43.9	54.0	62.2	70.1	77.2	78.2	70.4	60.7	50.6	37.7	(33)
(34)		MADB	35.4	39.9	49.0	60.2	69.3	76.2	81.0	82.8	75.7	66.1	54.3	40.8	(34)	
(35)	Mean Daily Temperature Range		MDBR	14.4	15.0	16.3	18.5	19.2	16.0	11.2	12.9	18.0	19.5	15.2	13.5	(35)
(36)		5% DB	MADB	17.6	19.6	22.3	23.8	25.0	21.0	15.4	15.6	20.5	23.0	19.9	17.3	(36)
(37)			MCWBR	14.2	14.6	14.5	13.3	12.1	9.3	8.1	7.8	10.6	13.1	14.5	13.6	(37)
(38)			MADB	16.5	17.1	20.1	19.3	19.8	15.1	11.8	12.8	16.2	19.4	17.6	16.2	(38)
(39)		5% WB	MCWBR	13.9	13.8	14.7	12.6	11.6	8.1	7.4	7.4	9.9	13.0	14.3	13.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.438	0.496	0.557	0.613	0.630	0.714	0.686	0.590	0.519	0.531	0.510	0.447	(40)	
(41)		taud	1.938	1.828	1.730	1.664	1.680	1.588	1.691	1.902	1.998	1.849	1.849	1.927	(41)	
(42)		Ebn at Noon	218	221	223	220	220	202	206	224	232	208	192	203	(42)	
(43)		Edh at Noon	44	56	68	77	77	84	75	60	51	54	48	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	717	1004	1390	1606	1769	1624	1369	1472	1407	1126	746	620	(44)	
(45)		RadStd	56	75	113	113	164	187	184	178	121	75	83	42	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+187	+124

Nomenclature: See separate page